

# **ANTIPSYCHOTIC DRUGS (NEUROLEPTICS)**

**By  
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# Antipsychotics Drugs ( Neuroleptics)

**Psychosis:** - Severe mental disorders  
- Delusions( false belief)  
- Hallucinations( false perceptions)

**Psychotic disorders:**

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- Schizophrenia
- Manic phase of bipolar (manic-depressive) illness
- Acute idiopathic psychotic illnesses
- Conditions marked by severe agitation



# What Is Schizophrenia?

- Schizophrenia is a psychological disorder
- It is classified separately from other disorders because it is not easily categorized as an anxiety or mood disorder
- Schizophrenia is characterized by the inability to separate reality from and a non-reality
- Schizophrenia patients often experience non-existent stimuli that create perceptions of things that do not exist, such as voices
- Symptoms may include: flat affect (exhibiting no emotion), delusions, hallucinations, or disorganized thinking



Where are those  
voices coming from?

The Radio Told Me To  
Free All The Zoo  
Animals

I Saw Elephants  
Under My Bed

Aliens are contacting me

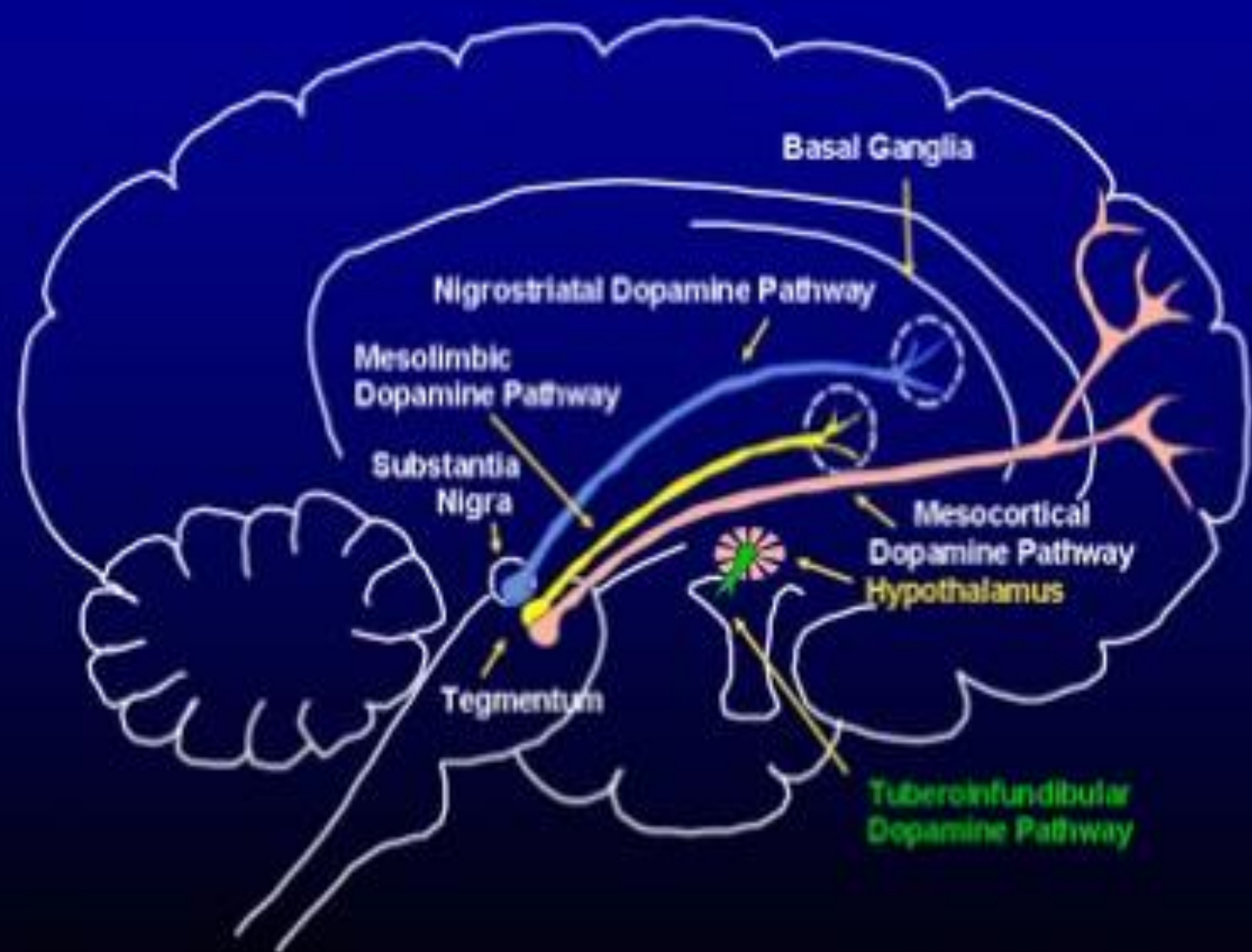
Maybe I  
Am Jesus

They're Following Me!

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# The Dopaminergic Pathways of the Brain

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# Dopamine Hypothesis

A01 /  
A02

## Biological Evidence

If inhibiting dopamine reduces schizophrenic symptoms what does that tell us about the cause of the disorder?

Phenothiazines  
(*anti-psychotics*) act to inhibit dopamine and reduce the symptoms of the disorder

L-Dopa increases dopamine levels and can produce many of the symptoms of schizophrenia

L-Dopa is given to sufferers of Parkinson's disease  
(*which is caused by low levels of dopamine*)

## 1<sup>st</sup> Generation (Neuroleptic):

Drugs that have experimental and clinical evidence of antagonism of D<sub>2</sub>-dopamine receptor activity, with substantial risk of adverse extrapyramidal effects (movement disorders) and increased release of prolactin.

EPS is more with Haloperidol (high potency).

EPS is less with Chlorpromazine (low potency).

## 2<sup>nd</sup> Generation (Atypical Antipsychotic):

Agents that are associated with substantially lower risks of such extrapyramidal effects.

Block both serotonin and dopamine receptors.

Higher risk of Diabetes, Weight gain and Hypercholesterolemia.

1<sup>st</sup> line therapy for schizophrenia.

Clozapine consider for refractory schizophrenia (bone marrow suppression (agranulocytosis), Seizures, Orthostasis).

Select the antipsychotic drug associated with more EPS?

- A) Clozapine
- B) Olanzapine
- C) Haloperidol
- D) Risperidone

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## 1<sup>ST</sup> GENERATION

### A) HIGH POTENCY

- FLUPHENAZINE
- THIOTHIXENE
- HALOPERIDOL
- PROCHLORPERAZINE
- PIMOZID

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### B) LOW POTENCY

- THIORIDAZINE
- CHLORPROMAZINE

## 2<sup>ND</sup> GENERATION ATYPICAL ANTIPSYCHOTICS

Loxapine, Clozapine (medium potency), Olanzapine (high potency), Quetiapine (low potency), Risperidone (high potency), Ziprasidone, Aripiprazole (high potency)

## **FIRST LINE AGENT**

- **2<sup>ND</sup> GENERATION'**
- **LOW EPS**
- **HIGH EFFICACY**

## **DISADVANTAGES OF 2<sup>ND</sup> GENERATION I.E CLOZAPINE**

- **DIABETES**
- **HYPERCHOLESTEROLEMIA**
- **AGRANULOCYTOSIS**
- **ORTHOSTATIC HYPOTENSION**
- **WEIGHT GAIN**

# Pharmacodynamics

- Prototype Typical antipsychotic  Chlorpromazine, Haloperidol
- Prototype Atypical antipsychotic  Clozapine

## A. DOPAMINERGIC SYSTEMS

### 1. Dopamine ( D2 receptors) antagonist (All first generation)

Typical Antipsychotics effective at occupancy level of 60% (D2).

All first generation and most of 2<sup>nd</sup> generation block D2 receptors in brain.

### 2. Serotonin receptor blocking Activity (5-HT<sub>2A</sub>)

Atypical Antipsychotics effective at occupancy levels of 40% - 50% (D2)

Most of these 2<sup>nd</sup> generation block serotonin (5-HT<sub>2A</sub>) receptors in the brain and periphery.

## **CLOZAPINE:**

**D<sub>1</sub>, D<sub>4</sub>, 5-HT-<sub>2</sub>, MUSCARINIC, ALFA ADRENERGIC,  
WEAK D<sub>2</sub> ANTAGONIST**

## **RISPERIDONE/OLANZAPINE:**

**STRONG 5HT-<sub>2A</sub>, WEAK D<sub>2</sub>**

## **ARIPIPRAZOLE:**

**PARTIAL AGONIST AT D<sub>2</sub> AND 5HT-<sub>1A</sub> RECEPTORS,  
ANTAGONIST AT 5HT<sub>2A</sub> RECEPTORS**

## **QUETIAPINE:**

**WEAK ANTAGONIST OF D<sub>2</sub> THAN 5HT<sub>2A</sub>**

## **PIMAVANSERIN:**

**INVERSE AGONIST AND ANTAGONIST AT 5HT<sub>2A</sub> & AF  
5HT<sub>2C</sub> RECEPTORS (Used in psychosis associated with  
Parkinson's disease)**

## Select the mechanism of action of risperidone

- a) Weak 5HT<sub>2A</sub> strong D<sub>2</sub> antagonist
- b) Block Alfa receptor
- c) Stimulate GABA receptor
- d) Strong 5HT<sub>2A</sub> weak D<sub>2</sub> antagonist

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Which antipsychotic drug is a partial agonist at dopamine D<sub>2</sub> receptors?

- A. Haloperidol
- B. Aripiprazole
- C. Chlorpromazine
- D. Clozapine

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Haloperidol  
Fluphenazine  
Thiothixine

More D<sub>2</sub> blockage



Increase risk of  
extrapyramidal  
side effects

Olanzapine  
Clozapine  
Quetiapine  
Aripiprazole  
Ziprasidone

Less D<sub>2</sub> blockage



Decrease risk of  
extrapyramidal  
side effects

- Atypical antipsychotic agents are at least as potent in inhibiting 5-HT<sub>2</sub> receptors as they are in inhibiting D<sub>2</sub> receptors

# ACTIONS

## 1. ANTIPSYCHOTIC EFFECTS

REDUCE HALLUCINATION AND DELUSION CALLED POSITIVE SYMPTOMS (D<sub>2</sub> BLOCKAGE IN MESOLIMBIC PATHWAY)

APATHY, BLUNTED AFFECT, IMPAIRED ATTENTION, COGNITIVE IMPAIRMENT ARE NEGATIVE SYMPTOMS NOT IMPROVED WITH FIRST GENERATION, LITTLE IMPROVED WITH 2<sup>ND</sup> GENERATION (CLOZAPINE)

## 2. EPS (D<sub>2</sub> BLOCKAGE AT NS PATHWAY BY FIRST GENERATION)

DYSTONIAS, PARKINSON LIKE SYMPTOMS, AKATHISIA, TARDIVE DYSKINESIA (D<sub>2</sub> BLOCKADE IN NIGROSTRIATAL PATHWAY IS THE CAUSE)

## 3. ANTIEMETIC (D<sub>2</sub> BLOCKADE IN CRTZ)

ALL EXCEPT ARIPIPRAZOLE

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## 4. ANTICHOLINERGIC

THIORIDAZINE, CHLORPROMAZINE, CLOZAPINE  
OLANZAPINE

BLURRED VISION, DRY MOUTH (Except clozapine that increases salivation), CONSTIPATION, URINE RETENTION, CONFUSION.

Which adverse effect of antipsychotic drugs is mainly due to dopamine D<sub>2</sub> receptor blockade in the nigrostriatal pathway?

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- A. Hyperglycemia
- B. Extrapyramidal symptoms
- C. Agranulocytosis
- D. Orthostatic hypotension

Which antipsychotic drug has prominent antimuscarinic effects?

- A. Clozapine
- B. Haloperidol
- C. Fluphenazine

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## OTHER EFFECTS:

- ☐ ORTHOSTATIC HYPOTENSION ,  
LIGHTHEADEDNESS ( ALFA ADRENERGIC  
RECEPTOR BLOCKADE)
- ☐ POIKILOOTHERMIA (ALTERED BODY TEMP)
- ☐ HYPERPROLACTINEMIA (D<sub>2</sub>- BLOCKADE)
- ☐ SEDATION (HI BLOCKADE)---Chlorpromazine,  
Clozapine, Olanzapine, Quetiapine
- ☐ SEXUAL DYSFUNCTION
- ☐ WEIGHT GAIN (2<sup>nd</sup> generation)

Which antipsychotic drug is particularly associated with hyperprolactinemia?

- A. Fluphenazine
- B. Clozapine
- C. Quetiapine
- D. Aripiprazole

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# Adverse Effects

## NEUROLOGICAL

Imbalance between inhibitory influence of dopamine (deficient) and excitatory influence (increased) of acetylcholine in the striatum

Less Dopaminergic VS more Cholinergic

- **Acute adverse extrapyramidal effects (high-potency D<sub>2</sub> dopamine receptor antagonists)**

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**Acute dystonia**

**Akathisia** (treated with propranolol/Benzodiazepines)

**Parkinsonism** (Bradykinesia, tremors, m.rigidity) (treated with Benztropine)

**Tardive dyskinesia** (treated with Val-benazine/Deutetra-benazine)

**Neuroleptic malignant syndrome**

## NEUROLEPTIC MALIGNANT SYNDROME

- Excessively rapid blockade of postsynaptic dopamine receptors.
- Occurs in patients who are extremely sensitive to the extrapyramidal effects of antipsychotic agents

Characterized by

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- High fever
- Muscle rigidity
- Altered mental status
- Stupor
- **Treatment : antiparkinsonism drugs( bromocriptine), diazepam, dantrolene.**

## **OTHERS**

**Dry mouth, Blurred vision, Constipation, Urinary retention**

**Orthostatic hypotension**

**Major ventricular arrhythmias (Ziprasidone/Thioridazine causes QT prolongation)**

**Weight gain**

**Hyperprolactinemia (Gynecomastia, Galactorrhea, Infertility, amenorrhea, Erectile dysfunction)**

**Agranulocytosis( clozapine)**

**Hyperlipidemia**

**Seizure (avoid antipsychotics in seizure disorders)**

# CLINICAL PHARMACOLOGY OF ANTIPSYCHOTIC AGENTS

## A. PSYCHIATRIC INDICATIONS

- Schizophrenia
- Schizoaffective disorders (Paliperidone)
- The manic phase in bipolar affective disorder
- Bipolar depression (Quetiapine, Lurasidone)
- Tourette's syndrome (Pimozid ,Risperidone, Haloperidol,)
- Autism associated with disrupted behavior (Risperidone/Aripiprazole)
- Refractory Depression (Quetiapine, Aripiprazole)

## B. NONPSYCHIATRIC INDICATIONS

- Prochlorperazine as antiemetic.
- Chlorpromazine Intractable hiccups
- Butyrophenone Droperidol is used in combination with an opioid fentanyl in neuroleptanesthesia

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## BRAIN WARM UP QUIZ

Q1. The primary mechanism of action of typical antipsychotic drugs is blockade of?

Q2. Which antipsychotic drug is most strongly associated with agranulocytosis?

Q3. Which antipsychotic drug is considered an atypical antipsychotic?

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Q4. Neuroleptic malignant syndrome is most strongly associated with blockade of which receptor?

Q5. Which adverse effect is particularly associated with chlorpromazine?

Q6. Which antipsychotic drug has a relatively low risk of extrapyramidal symptoms?

Q6. Which antipsychotic drug is useful in treatment-resistant schizophrenia?

Q7. Which antipsychotic drug has a relatively low propensity to cause extrapyramidal symptoms because of its partial dopamine agonist activity? R.FAROOQUI

Q8. The antipsychotic drug most strongly associated with QT interval prolongation among the following is?

Q9. Which antipsychotic drug is commonly used for the treatment of Tourette syndrome because of its dopamine D<sub>2</sub> receptor blockade?

Q10. The therapeutic effect of antipsychotic drugs is primarily related to blockade of dopamine receptors in which pathway?

## OPTIONS

- a) CLOZAPINE
- b) D<sub>2</sub> RECEPTORS
- c) D<sub>1</sub> RECEPTORS
- d) DOPAMINE RECEPTOR
- e) QUETIAPINE
- f) HALOPERIDOL
- g) ORTHOSTATIC HYPOTENSION
- h) ARIPIPRAZOLE
- i) Thioridazine
- j) Mesolimbic pathway

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